NEST NEPAL

Nest Nepal Business Solutions
Kupondole, Lalitpur

Backend Developer Intern

Intern Task

Backend Developer Intern · Take-home coding assignment

Mini SMS Gateway

Accept. Charge. Send. Fail over.

06	05	4–5	Any
Core requirements	Bonus items	Days	Stack

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THE BRIEF

Task Overview

You are asked to build the core of **Nest SMS**, our bulk SMS platform, as a REST API. It accepts SMS requests, charges credits and sends them through mock providers, **switching to another provider when one fails**.

AREA	WHAT IT COVERS
Auth and wallet	JWT login, one credit balance per user
Mock providers	2–3 fake SMS routes with priority, balance and failure rate
Sending	Validation, cost, atomic credit deduction, failover, refunds
Reporting	Message logs and route health

REQUIRED OUTPUT

REST API

Any language or framework, with a database that supports transactions.

CORE CHALLENGE

Never lose or double-spend a credit.

Two parallel requests, a failing route or a crash must never leave a wallet wrong.

Ground rules

- ✓ **Any stack.** Use any language or framework you are comfortable with.
- ✓ **AI tools are fine.** You must be able to explain every line in the follow-up interview.
- ✓ **Core first.** Most of the score comes from the core requirements. Bonus items are optional.

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SHARED RULES

Counting SMS

Every role uses the same rules for counting SMS parts and checking phone numbers. Get these right first, because the cost of every message depends on them.

ENCODING	WHEN	1 SMS HOLDS	LONGER MESSAGES
GSM-7	Only standard characters (A–Z, 0–9, common symbols)	160 characters	Parts of 153
Unicode	Any other character, such as Nepali (Devanagari) or emoji	70 characters	Parts of 67

Phone numbers

A valid Nepali mobile number is **10 digits starting with 97 or 98**. An optional `+977` or `977` prefix is allowed and must be normalised to the 10-digit form.

INPUT	RESULT	WHY
`9841234567`	`9841234567`	Valid as it is
`+9779801122334`	`9801122334`	Prefix removed
`9779761234567`	`9761234567`	Prefix removed
`9612345678`	invalid	Does not start with 97 or 98
`984123456`	invalid	Only 9 digits

✓ **Example:** a 200-character English message is 2 parts (153 + 47). A 100-character Nepali message is also 2 parts (67 + 33). One emoji in an English message makes the whole message Unicode.

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PART A · REQUIRED

Core Requirements

DIRECTION

Build an API that takes a send request and gets every message out, or gives the credits back.

01	**Auth**	Register and log in with JWT. Each user has a wallet with a `credits` balance.
02	**Mock providers**	2–3 fake providers (for example `ROUTE_A`, `ROUTE_B`). Each has a `priority`, its own balance, a configurable failure rate (ROUTE_A fails 30% of the time) and a `getBalance()` method.
03	**`POST /sms/send`**	Body: `{ to: string[], message }`. • Validate and normalise numbers. Reject invalid ones clearly. • Calculate the parts per message and the total cost. • Reject the request if credits are not enough. • Deduct credits in a DB transaction, so parallel requests cannot overspend. • Send via the top-priority provider with balance. On failure, try the next.
04	**`GET /sms/logs`**	Paginated history, filterable by status (`pending`, `sent`, `failed`), date and number.
05	**`GET /routes/health`**	For each route: balance, success rate over the last 24 h, and a health %. Use your own formula and explain it in the README.
06	**Refunds**	Give the credits back for messages that fail on every route.

✓ **Watch for:** fire 20 parallel send requests at a wallet that can only afford 10. Exactly 10 should succeed.

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PART B · BONUS

Going Further

05

DIRECTION

Optional. Pick the ones that make your gateway closer to production.

B1	**Queue**	BullMQ or Redis, so sending happens in the background and the API responds immediately.
B2	**Rate limiting**	Per user, for example 60 requests a minute.
B3	**Delivery reports**	A webhook endpoint that updates a message's status.
B4	**Unit tests**	For the segment counter, number validation and failover.
B5	**Docker Compose**	One command to run the API, database and queue.

✓ **Watch for:** a solid Part A with two bonus items is better than everything done in a hurry.

Deliverables

What to hand in when you are done.

DELIVERABLE	FORMAT	REQUIREMENT
Source code	GitHub repository	The API, migrations or schema, and seed data for providers.
README	Markdown in the repo	Setup steps, decisions you made, your health formula, and what you would improve.
Demo	3–5 min recording or deployed link	Show a send, a failover, a refund and the logs.

✓ Before you submit

- ✓ Numbers with `+977` or `977` are normalised. Invalid numbers get a clear error.
- ✓ Parts and cost match the shared SMS rules for GSM-7 and Unicode.
- ✓ Credits are deducted inside a transaction. Parallel requests cannot overspend.
- ✓ A failed route falls over to the next provider by priority.
- ✓ Messages that fail on every route are refunded.
- ✓ Logs filter by status, date and number, and are paginated.
- ✓ The README explains your route health formula.
- ✓ The project runs from the README steps on a fresh machine.

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All the Best

Show us an API that never loses a credit.

Submission | Notes

FOCUS
Correct transactions and failover matter more than extra endpoints.

SUGGESTED TIME
4 to 5 days. Most of the score comes from the core requirements.

HOW TO SUBMIT
Reply to the email this brief came with: your GitHub repo link plus a 3–5 minute screen recording or a deployed link.

LOCATION
2F, Allure Complex, Kupondole, Lalitpur, Nepal